



IMT Atlantique

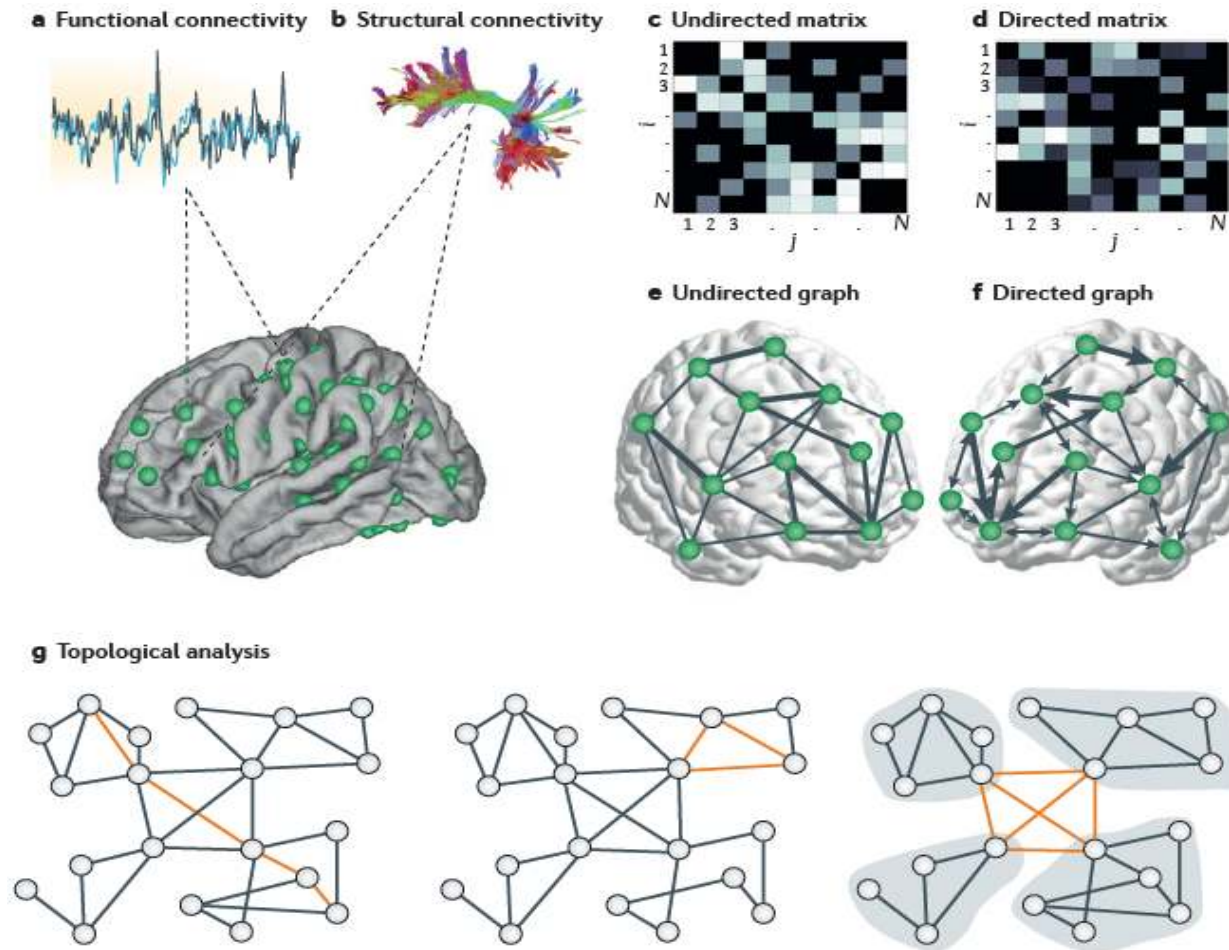
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EVALUATING GRAPH SIGNAL PROCESSING FOR NEUROIMAGING THROUGH CLASSIFICATION AND DIMENSIONALITY REDUCTION

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NEUROIMAGING AND NETWORK SCIENCE



Fornito et al. 2015
Nature Neuro

The application of Graph Theory for neuroimaging is now widespread in the neuroimaging community.

WHY GRAPH SIGNAL PROCESSING ?

- ▶ Graph Signal Processing (GSP) attempts to generalize univariate signal processing (eg Fourier analysis) to irregular domains
- ▶ Graph Laplacian :

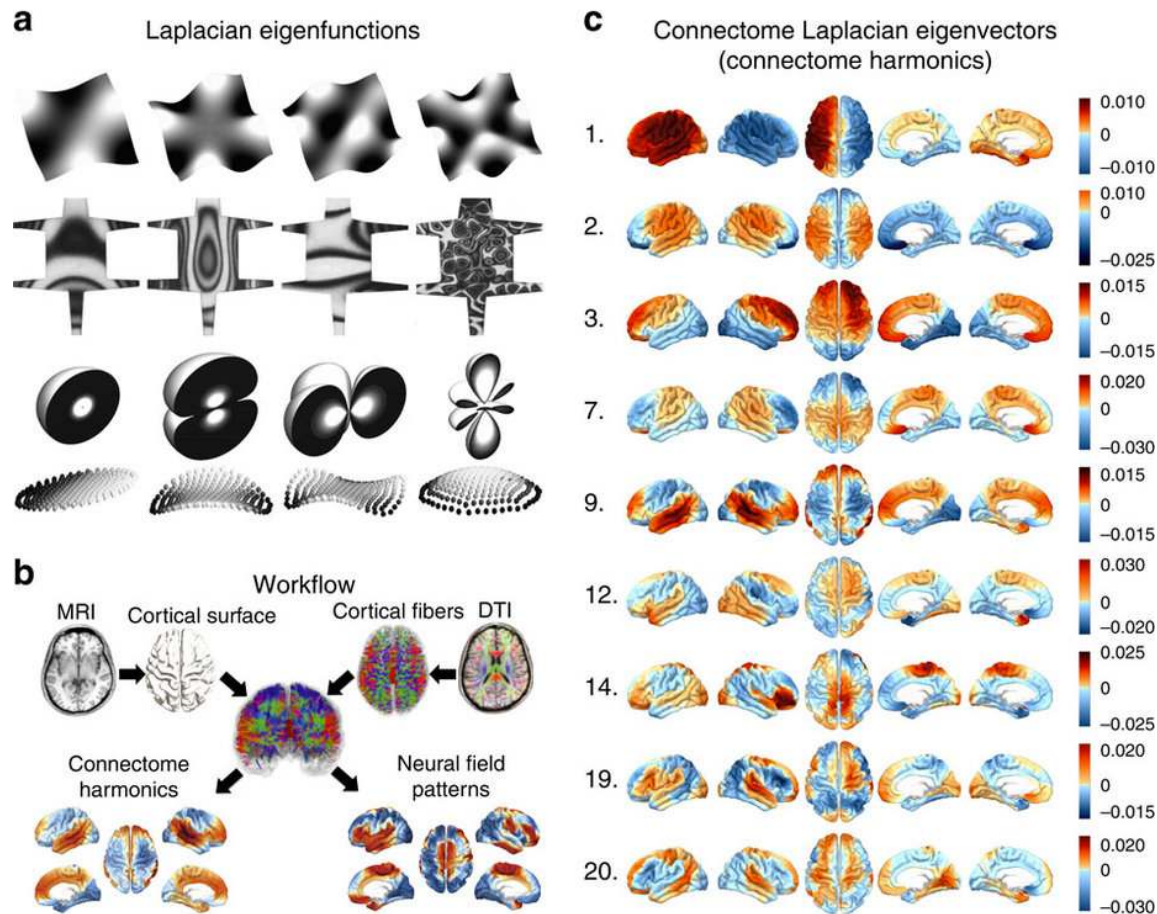
$$L = D - A$$

A : Adjacency matrix, D : degree matrix

- ▶ Graph Signals are a mapping of a signal on the graph nodes
- ▶ Graph Fourier Transform exploits the eigenvectors of L (harmonics)
- ▶ Classical signal processing can be obtained by taking a ring graph
- ▶ Applications of GSP
 - Decomposition of signals in the Graph Fourier domain
 - Filtering
 - ...
- ▶ Properties of the graph Laplacian for brain graphs

THE NEXT STEP ?

- Laplacian eigenvectors of a graph based on brain structure (gray and white matter) can predict spontaneous brain activity (Atasoy, Donnelly & Pearson, 2017, Nature communications)



THE NEXT STEP ?

- Laplacian eigenvectors can be interpreted functionally (Margulies et al. 2016, PNAS. here termed “connectivity gradients”)

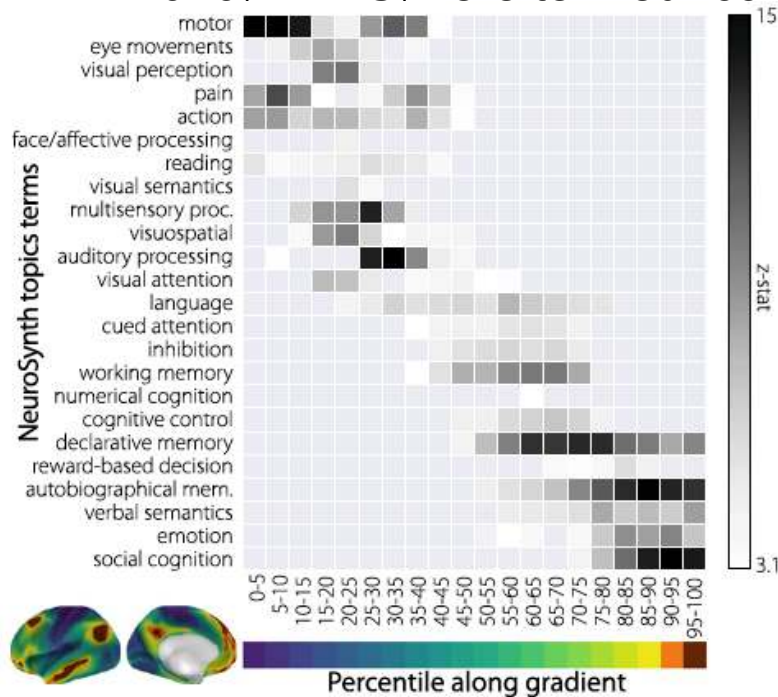
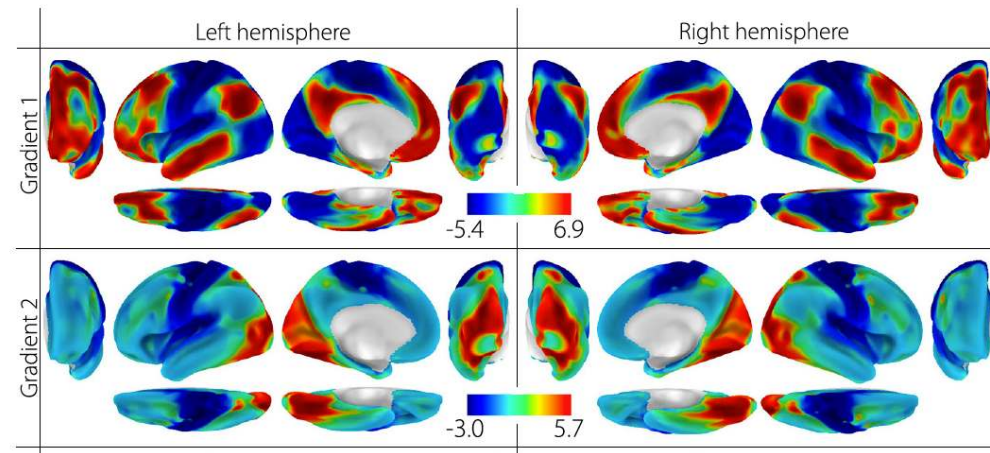


Fig. 4. NeuroSynth metaanalysis of regions of interest along the principal gradient using 24 topic terms. Terms are ordered by the weighted mean of their location along the gradient. Sensory processing terms are located at the top followed by domain-general cognitive functions and then, higher-order abstract cognitive and memory-related processes. Similar results using the BrainMap database are available in [SI Materials and Methods](#). autobiographical mem., autobiographical memory; multisensory proc., multisensory processing.



-> Therefore GSP appears as an ideal framework to analyse, predict and interpret brain activity.

GSP APPLIED TO NEUROIMAGING

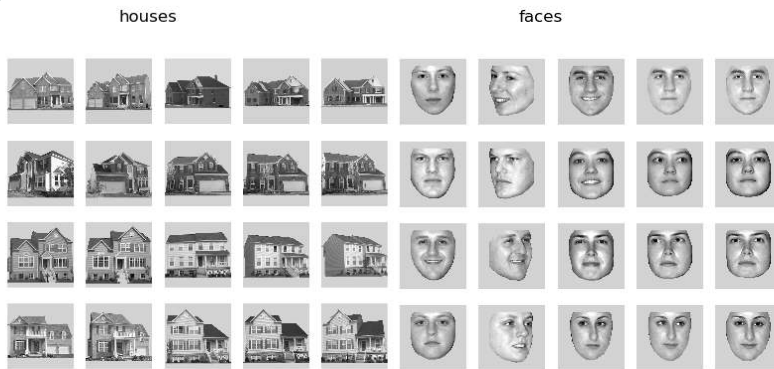
- ▶ Graph Frequencies + PCA (Huang et al. 2016) : statistical analysis on a motor learning task and fMRI data.
- ▶ Dimensionality reduction (Rui et al. 2016) by learning subspaces on graph frequencies on MEG data.
- ▶ Denoising and dimensionality reduction on EEG data using graph laplacian – SPHARA (Graichen et al. 2015)
- ▶ Low rank estimation / denoising (Liu et al. 2016)
- ▶ Measure derived from graph signal smoothness combined with graph modularity (Modular Dirichlet Energy) to analyse an EEG experiment (Smith et al 2017)

OUR CONTRIBUTION

- ▶ Goal: use GSP for supervised classification of brain activity (fMRI)
- ▶ Key Questions
 - Can GSP improve classification accuracy?
 - What is the influence of the graph?
 - Can GSP be exploited for dimensionality reduction in the context of brain data?
- ▶ Experiments
 - Simulated fMRI data
 - Haxby dataset

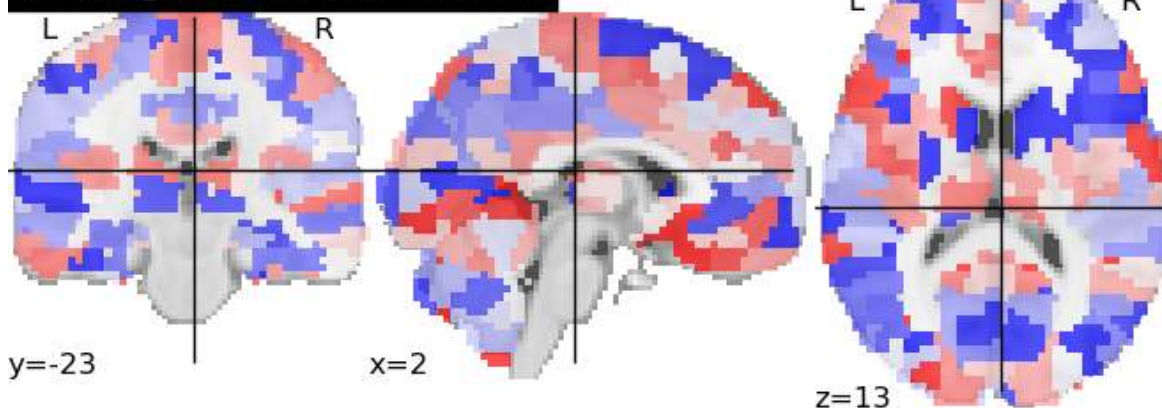
All results are presented here using classification with linear support vector machines.

Haxby et al. 2001, Science



Brain parcellation to create regions of interests (ROIs)
Geometry : baricenter of ROIs

444 regions of brain clusters



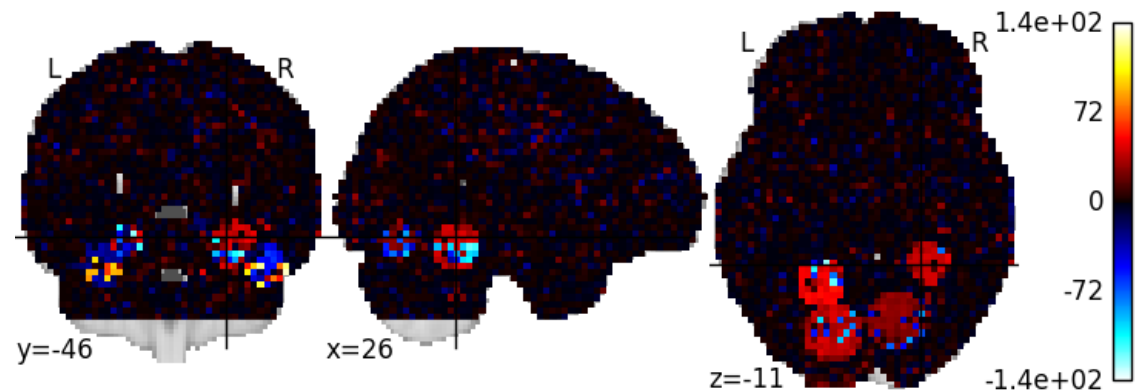
BASC Atlas,
Bellec et al.
Neuroimage 2010

Simulated activity in 6 areas in two conditions (Faces Vs Houses):

- Bilateral Primary Visual Cortex V1 (similar activation for both conditions),
- Bilateral Fusiform Face Area (stronger activation for faces)
- Bilateral Parahippocampal Place area (stronger activation for houses).

Simulated noise on spatial locations and activation amplitude.
Simulations were grouped in “Easy” or “Difficult” cases depending on noise.

Contrast Condition 1 – Condition 2



DIFFERENT TYPES OF GRAPHS

- ▶ Graphs based on 3D coordinates
 - Weights = gaussian kernel on distances between ROI coordinates
 - All weights (Full) or Thresholded (Geometric)

- ▶ Functional graphs
 - Weights = statistical measures of similarity between ROI signals
 - Absolute values of Correlation or Covariance

- ▶ Graph based on smoothness priors (Kalofolias)

- ▶ Mixed graphs
 - Thresholded geometric support + covariance weights
 - Fundis = product of Full and Covariance

- ▶ N.B. Graph Fourier Transforms are isometries

From initial data :

- ▶ PCA or ICA
 - Select k first components
- ▶ K-best features selected by Analysis of Variance (k-best ANOVA)
 - Perform all univariate ANOVA comparing conditions
 - Select k nodes that discriminate best the conditions

From low (LF) and high frequencies (HF) in the GFT domain :

- ▶ Select k first (resp. last) eigenvalues of L to get GFT LF (resp. GFT HF)
- ▶ Graph Sampling
 - Calculate the graph weighted coherence for LF and HF
 - Extract k nodes concentrating most energy

SIMULATED DATA – 50 COMPONENTS

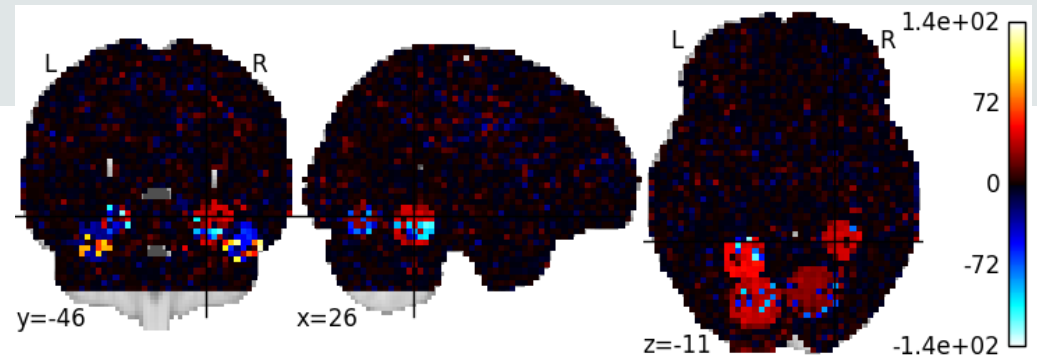
TABLE I
ACCURACY OF THE CLASSIFICATION FOR ALL GRAPH-BASED SAMPLING METHODS (IN %) FOR THE TWO DIFFICULTY GROUPS. STARRED NUMBERS INDICATE BEST SCORES IN THEIR CATEGORY.

Graph Types	GFT LF	GFT HF	GFT ANOVA	GS LF	GS HF
Difficult					
Full	54.8%	51.1%	66.0%	52.0%	51.3%
Geometric	56.7%	64.8%	64.8%	50.5%	65.2%
Correlation	52.4%	66.8%	64.7%	50.9%	60.3%
Covariance	52.4%	67.6%	65.2%	51.2%	66.2%
Kalofolias	61.6%	51.9%	65.9%	61.6%	51.9%
Semilocal	53.8%	69.5%	65.6%	50.3%	72.5% *
Fundis	54.9%	64.2%	65.1%	49.7%	62.8%
Easy					
Full	65.1%	60.0%	88.9%	49.6%	60.6%
Geometric	71.3%	79.4%	86.0%	57.8%	79.1%
Correlation	59.5%	86.5%	86.9%	53.5%	75.2%
Covariance	57.2%	87.0%	88.8%	52.8%	84.8%
Kalofolias	88.2%	54.2%	89.4%	87.5%	52.6%
Semilocal	61.3%	87.9%	88.6%	52.3%	90.9% *
Fundis	67.4%	77.5%	86.7%	52.4%	77.5%

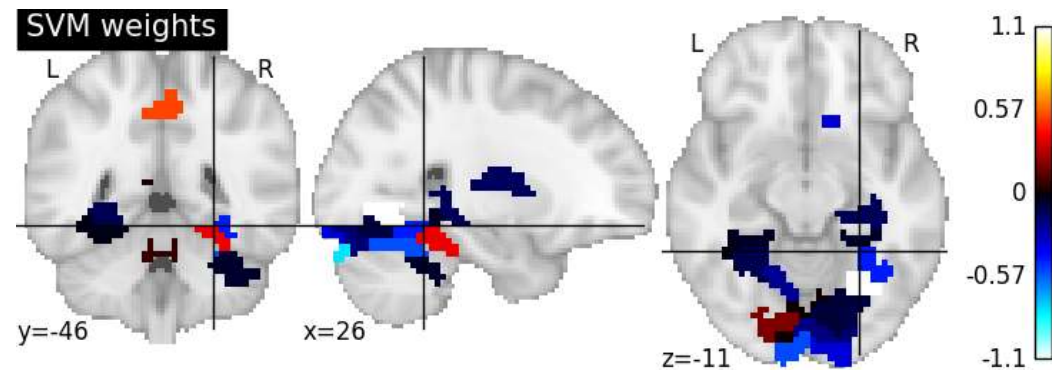
-> next results are shown only for the semilocal graph

RESULTS USING SIMULATED DATA

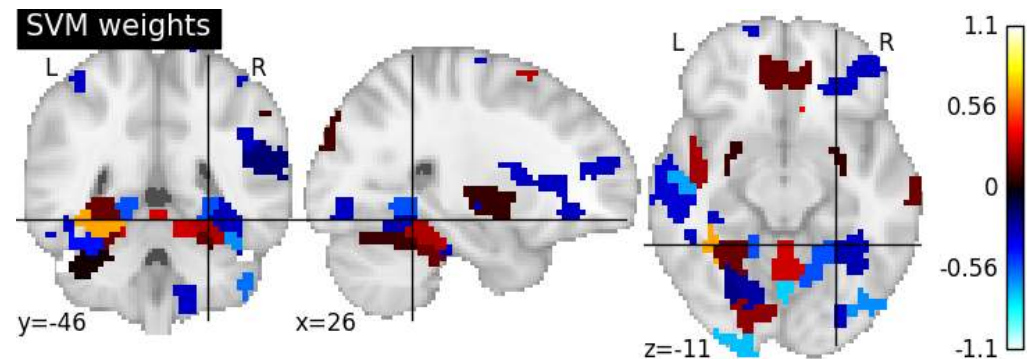
Contrast Condition 1 – Condition 2



SVM weights after
GS



SVM weight after
ANOVA feature selection



RESULTS SIMULATION AND HAXBY SEMILOCAL GRAPH

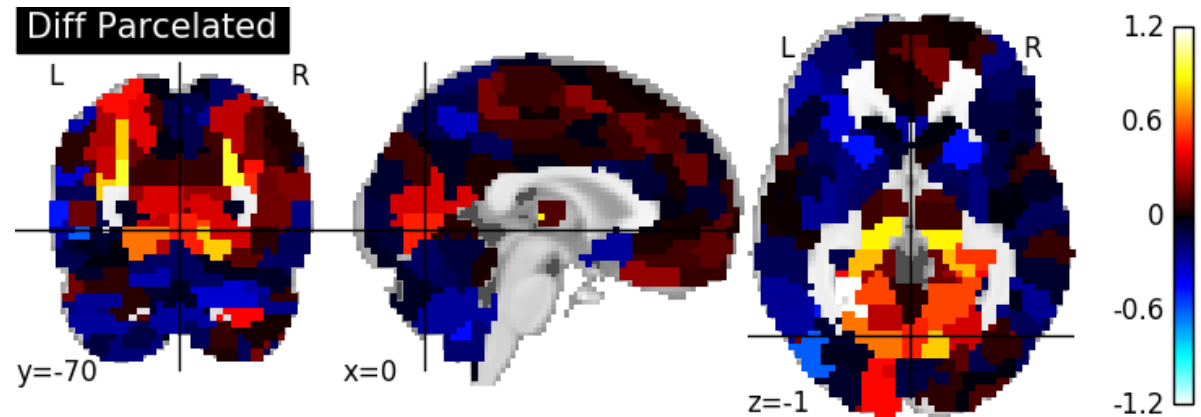
TABLE II
COMPARISON OF GRAPH SAMPLING (*Semilocal* GRAPH), PCA, ICA AND ANOVA. CLASSIFICATION ACCURACY WITH 50 COMPONENTS FOR THE SIMULATED AND HAXBY DATASETS.

Method	Simulation		Haxby	
	Easy	Difficult	Face-House	Cat-Face
PCA	88.8%	65.5%	82.7%	63.6%
ICA	90.2%	65.3%	84.4%	67.0%
ANOVA	92.1%	67.3%	85.5%	65.5%
Graph sampling	90.9%	72.5%	88.2%	69.0%

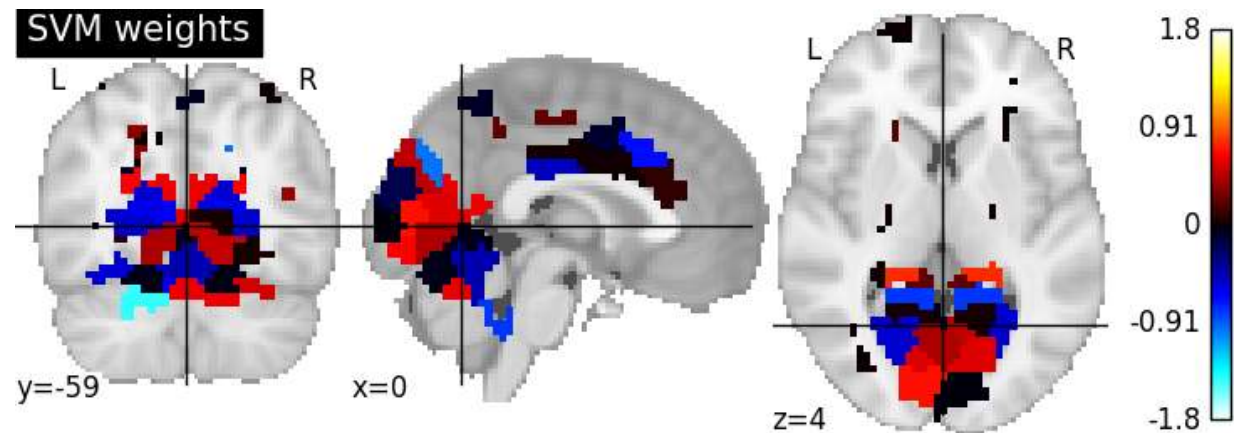
RESULTS ON HAXBY DATASET

Dim: 50

Contrast House-Face



SVM weight after
Graph Sampling
(Semilocal)



- ▶ Dimensionality reduction combined with GSP is a promising avenue
- ▶ Classification
 - On simulated data, GSP using semi-local graph and graph sampling yields significant performance improvements in a difficult scenario
 - On a real dataset : when keeping few dimensions, GSP + semilocal and graph sampling may improve classification accuracy
- ▶ Perspectives
 - Use a structural graph from dMRI ?
 - Role of classification technique ? (linear SVC used so far)

Questions?

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